

Hylogical Conjugacy: A Bipartite Algebra for Local Transforms and Paired Optimisation

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Abstract

This research note extends a local physical–semantic vectorisation to transformations. Given a local correspondence between observed physical equivalence classes and a semantic vector space, quotient-compatible physical transformations induce semantic calculations by conjugation. The correspondence preserves composition. It also preserves identities and inverses where defined. Local dynamics conjugate under the same correspondence. Tangent operator algebras, metrics, and scalar observables admit corresponding descriptions. The result is substantive only when the physically admissible processes and semantic calculations are specified independently. Ordinary conjugation yields an equivalence, not a categorical duality. A duality requires an additional order-reversing correspondence.

1 Setup

A prior construction [1] obtains, on a regular local domain, a semantic vectorisation

$$h_U : \mathcal{P}_U \longrightarrow \mathcal{V}_U, \quad (1)$$

where

$$\mathcal{P}_U := \pi_E^{-1}(U), \quad \mathcal{V}_U := V_{\text{int},U}, \quad (2)$$

and h_U is a diffeomorphism. A point $p \in \mathcal{P}_U$ is an observed physical equivalence class rather than a unique complete physical configuration.

For concision write

$$h := h_U. \quad (3)$$

Definition 1 (Semantic Vector-Space Calculation). A local semantic vector-space (SVS) calculation is a map

$$T : \mathcal{V}_U \longrightarrow \mathcal{V}_U. \quad (4)$$

Definition 2 (Physical Quotient Transformation). A physical quotient transformation is a map

$$F : \mathcal{P}_U \longrightarrow \mathcal{P}_U. \quad (5)$$

The question is whether calculations in $\mathcal{V}_U$ correspond with physically admissible transformations in $\mathcal{P}_U$.

2 Physical Descent

Take a fixed observation window B . Recall the physical equivalence relation

$$s_1 \sim_B s_2 \iff s_1|_B = s_2|_B. \quad (6)$$

Consider a physical process

$$\hat{F} : \Gamma(P, S) \longrightarrow \Gamma(P, S). \quad (7)$$

Definition 3 (Quotient-Compatible Physical Process). The process $\hat{F}$ is quotient-compatible on B when

$$s_1 \sim_B s_2 \implies \hat{F}(s_1) \sim_B \hat{F}(s_2). \quad (8)$$

Lemma 1 (Physical Descent). Every quotient-compatible $\hat{F}$ induces a well-defined transformation

$$\overline{F}_B : Q_B(S) \longrightarrow Q_B(S), \quad (9)$$

defined by

$$\overline{F}_B([s]_B) := [\hat{F}(s)]_B. \quad (10)$$

Proof. If $[s_1]_B = [s_2]_B$, then $s_1 \sim_B s_2$. Quotient compatibility gives

$$\hat{F}(s_1) \sim_B \hat{F}(s_2), \quad (11)$$

and therefore

$$[\hat{F}(s_1)]_B = [\hat{F}(s_2)]_B. \quad (12)$$

Thus (10) is independent of representative.

The same condition applies fibrewise across a local U . Therefore, an underlying physical process yields a transformation of $\mathcal{P}_U$ whenever it respects the observational quotient.

Definition 4 (Physical Realisation of an SVS Calculation). An SVS calculation T has physical realisation F when

$$T \circ h = h \circ F. \quad (13)$$

Equivalently,

$$T = h \circ F \circ h^{-1}. \quad (14)$$

Thus, an SVS calculation has a physical counterpart whenever its h -conjugate is induced by a physically admissible quotient-compatible process.

3 Symmetric Transformation Structure

Let

$$\mathcal{M}_P \subseteq \text{End}(\mathcal{P}_U) \quad (15)$$

denote the admissible physical quotient transformations and

$$\mathcal{M}_V \subseteq \text{End}(\mathcal{V}_U) \quad (16)$$

denote the admissible semantic calculations.

Define the conjugation map

$$\mathcal{C}_h(F) := h \circ F \circ h^{-1}. \quad (17)$$

Lemma 2 (Composition Preservation). For $F_1, F_2 \in \mathcal{M}_P$,

$$\mathcal{C}_h(F_2 \circ F_1) = \mathcal{C}_h(F_2) \circ \mathcal{C}_h(F_1). \quad (18)$$

Furthermore,

$$\mathcal{C}_h(\text{id}_{\mathcal{P}_U}) = \text{id}_{\mathcal{V}_U}, \quad (19)$$

and, whenever F^{-1} exists,

$$\mathcal{C}_h(F^{-1}) = \mathcal{C}_h(F)^{-1}. \quad (20)$$

Proof. Substitute (17) and cancel adjacent $h^{-1} \circ h$.

Assumption 3.1 (Independent Admissibility). The collection $\mathcal{M}_P$ is specified by the governing physical laws, observer, and interaction protocol. The collection $\mathcal{M}_V$ is specified by the chosen semantic calculus. Neither collection is defined as the transported image of the other.

This condition distinguishes a physical–semantic claim from a coordinate identity.

Definition 5 (Physical–Semantic Symmetry Condition).

$$\mathcal{C}_h(\mathcal{M}_P) = \mathcal{M}_V. \quad (21)$$

Theorem 1 (Transformation Equivalence). Under the physical–semantic symmetry condition,

$$\mathcal{C}_h : \mathcal{M}_P \longrightarrow \mathcal{M}_V \quad (22)$$

is an isomorphism of transformation monoids.

Proof. Equation (18) preserves composition and (19) preserves identity. Bijectivity follows from h^{-1} , since

$$F = h^{-1} \circ T \circ h. \quad (23)$$

Thus the admissible physical and semantic transformations have equivalent composition structure.

Corollary 1 (Symmetric Impossibility). Under (21),

$$T \notin \mathcal{M}_V \iff h^{-1} \circ T \circ h \notin \mathcal{M}_P. \quad (24)$$

Therefore, physical impossibility and semantic inadmissibility correspond under the local equivalence.

4 Tangent Operator Algebra

The general transformation structure preserves composition but need not preserve linear addition because h need not itself be linear. A linear algebra obtains locally on tangent spaces.

Fix

$$p \in \mathcal{P}_U, \quad v = h(p). \quad (25)$$

Since h is a diffeomorphism,

$$J_h(p) := Dh|_p : T_p \mathcal{P}_U \longrightarrow T_v \mathcal{V}_U \quad (26)$$

is a linear isomorphism.

For a linear operator

$$A : T_p \mathcal{P}_U \longrightarrow T_p \mathcal{P}_U, \quad (27)$$

define

$$\mathcal{A}_h(A) := J_h(p) A J_h(p)^{-1}. \quad (28)$$

Theorem 2 (Local Operator-Algebra Equivalence). The map $\mathcal{A}_h$ preserves

$$\mathcal{A}_h(A + B) = \mathcal{A}_h(A) + \mathcal{A}_h(B), \quad (29)$$

$$\mathcal{A}_h(\lambda A) = \lambda \mathcal{A}_h(A), \quad (30)$$

and

$$\mathcal{A}_h(A_2 A_1) = \mathcal{A}_h(A_2) \mathcal{A}_h(A_1). \quad (31)$$

Thus

$$\text{End}(T_p \mathcal{P}_U) \cong \text{End}(T_v \mathcal{V}_U) \quad (32)$$

as associative operator algebras.

When the operators are bounded and $J_h(p)$ is a bounded linear isomorphism,

$$\sigma(\mathcal{A}_h(A)) = \sigma(A). \quad (33)$$

Hence transported eigenvalues, spectral degeneracies, and invariant subspaces have physical and semantic representatives.

5 Dynamical Equivalence

Let

$$\Phi_t : \mathcal{P}_U \longrightarrow \mathcal{P}_U \quad (34)$$

be a physically admissible local flow or semigroup. Define

$$\Psi_t := h \circ \Phi_t \circ h^{-1}. \quad (35)$$

Theorem 3 (Local Dynamical Conjugacy).

$$h \circ \Phi_t = \Psi_t \circ h. \quad (36)$$

If

$$\Phi_{t+s} = \Phi_t \circ \Phi_s, \quad (37)$$

then

$$\Psi_{t+s} = \Psi_t \circ \Psi_s. \quad (38)$$

Thus physical trajectories and semantic trajectories correspond under h .

If X_P generates Φ_t , the corresponding semantic vector field satisfies

$$X_V(h(p)) = Dh|_p X_P(p). \quad (39)$$

Consequently, fixed points map to fixed points. Local linear stability operators are related by the tangent conjugation in (28). Their conjugacy invariants therefore agree.

An SVS evolution is, under these conditions, a coordinate description of a quotient-physical evolution.

6 Metric and Observable Correspondence

Take independently specified metrics

$$g_P \quad \text{on } \mathcal{P}_U, \quad g_V \quad \text{on } \mathcal{V}_U. \quad (40)$$

Definition 6 (Metric Symmetry).

$$g_P = h^* g_V. \quad (41)$$

Explicitly, for $u, w \in T_p \mathcal{P}_U$,

$$g_{P,p}(u, w) = g_{V,h(p)}(Dh|_p u, Dh|_p w). \quad (42)$$

Under (41), h is a local isometry. Physical distinguishability and semantic distinguishability are then the same metric quantity in corresponding coordinates.

Now let

$$f_V : \mathcal{V}_U \longrightarrow \mathbb{R} \quad (43)$$

be a semantic observable. Its corresponding physical quotient observable is

$$f_P := f_V \circ h. \quad (44)$$

Conversely, an observable on complete physical states descends to the quotient only when

$$s_1 \sim_B s_2 \implies f_P(s_1) = f_P(s_2). \quad (45)$$

Thus semantic observables correspond only with physical observables insensitive to information discarded by the observational quotient.

7 Paired Optimisation

The preceding conjugacy permits optimisation problems to be compared across the physical and semantic descriptions. Let

$$L_P : P_U \longrightarrow \mathbb{R}$$

be a physical objective and let

$$L_V : V_U \longrightarrow \mathbb{R}$$

be a semantic objective.

Definition 1 (Optimisation Symmetry). *The pair (L_P, L_V) satisfies optimisation symmetry with respect to h when*

$$L_P = L_V \circ h.$$

Equivalently,

$$L_V = L_P \circ h^{-1}.$$

As with the transformation symmetry condition, the substantive case is one in which L_P and L_V are specified independently and the preceding equality is then established, rather than one objective being defined merely as the transported image of the other.

Theorem 1 (Correspondence of Optima). *Suppose that*

$$L_P = L_V \circ h.$$

Then, for every $p_\star \in P_U$ and

$$v_\star := h(p_\star),$$

the point $p_\star$ is a local minimiser of L_P if and only if $v_\star$ is a local minimiser of L_V . Likewise, local maxima correspond.

In particular,

$$h\left(\arg \min_{p \in P_U} L_P(p)\right) = \arg \min_{v \in V_U} L_V(v),$$

whenever the indicated global minima exist.

Proof. Since h is a diffeomorphism, it is in particular a homeomorphism. Thus neighbourhoods of $p_\star$ correspond bijectively with neighbourhoods of $v_\star = h(p_\star)$. Moreover,

$$L_P(p) = L_V(h(p)).$$

Hence

$$L_P(p_\star) \leq L_P(p)$$

throughout some neighbourhood of $p_\star$ if and only if

$$L_V(v_\star) \leq L_V(v)$$

throughout the corresponding neighbourhood of $v_\star$. The global statement follows from bijectivity of h . □

Corollary 1 (Correspondence of Critical Points). *If L_P and L_V are differentiable and*

$$L_P = L_V \circ h,$$

then

$$dL_P|_p = dL_V|_{h(p)} \circ Dh|_p.$$

Since $Dh|_p$ is invertible,

$$dL_P|_p = 0 \iff dL_V|_{h(p)} = 0.$$

Thus critical points correspond under h .

Now impose the metric symmetry condition of Section 6,

$$g_P = h^* g_V.$$

Theorem 2 (Gradient Correspondence). *Suppose*

$$L_P = L_V \circ h$$

and

$$g_P = h^* g_V.$$

Then the metric gradients satisfy

$$\text{grad}_{g_V} L_V(h(p)) = Dh|_p \text{grad}_{g_P} L_P(p).$$

Equivalently,

$$\text{grad}_{g_P} L_P(p) = Dh|_p^{-1} \text{grad}_{g_V} L_V(h(p)).$$

Proof. For every $u \in T_p P_U$,

$$dL_P|_p(u) = dL_V|_{h(p)}(Dh|_p u) \tag{1}$$

$$= g_{V,h(p)}(\text{grad}_{g_V} L_V(h(p)), Dh|_p u). \tag{2}$$

By

$$g_P = h^* g_V,$$

this equals

$$g_{P,p}(Dh|_p^{-1} \text{grad}_{g_V} L_V(h(p)), u).$$

Uniqueness of the metric gradient therefore gives

$$\text{grad}_{g_P} L_P(p) = Dh|_p^{-1} \text{grad}_{g_V} L_V(h(p)).$$

□

Theorem 3 (Conjugacy of Gradient Flows). *Let the physical optimisation dynamics satisfy*

$$\dot{p}(t) = -\text{grad}_{g_P} L_P(p(t)).$$

Define

$$v(t) := h(p(t)).$$

Under optimisation symmetry and metric symmetry,

$$\dot{v}(t) = -\text{grad}_{g_V} L_V(v(t)).$$

Hence the physical and semantic gradient flows are conjugate under h .

Proof. By the chain rule,

$$\dot{v}(t) = Dh|_{p(t)} \dot{p}(t).$$

Therefore

$$\dot{v}(t) = -Dh|_{p(t)} \text{grad}_{g_P} L_P(p(t)) \tag{3}$$

$$= -\text{grad}_{g_V} L_V(h(p(t))) \tag{4}$$

$$= -\text{grad}_{g_V} L_V(v(t)). \tag{5}$$

□

Consequently, corresponding optimisation trajectories have identical objective values:

$$L_P(p(t)) = L_V(v(t)).$$

Moreover,

$$\frac{d}{dt} L_P(p(t)) = - \left\| \text{grad}_{g_P} L_P(p(t)) \right\|_{g_P}^2,$$

while

$$\frac{d}{dt} L_V(v(t)) = - \left\| \text{grad}_{g_V} L_V(v(t)) \right\|_{g_V}^2.$$

Metric symmetry implies

$$\left\| \text{grad}_{g_P} L_P(p) \right\|_{g_P} = \left\| \text{grad}_{g_V} L_V(h(p)) \right\|_{g_V},$$

so the corresponding objective descent rates agree.

Definition 2 (Paired Optimisation Map). *Let*

$$\mathcal{O}_P : P_U \longrightarrow P_U$$

be an admissible physical optimisation update. Its h -paired semantic update is

$$\mathcal{O}_V := h \circ \mathcal{O}_P \circ h^{-1}.$$

For iterates

$$p_{k+1} = \mathcal{O}_P(p_k), \quad v_k = h(p_k),$$

one then has

$$v_{k+1} = \mathcal{O}_V(v_k).$$

Thus the optimisation diagram commutes:

$$\begin{array}{ccc} P_U & \xrightarrow{\mathcal{O}_P} & P_U \\ h \downarrow & & \downarrow h \\ V_U & \xrightarrow{\mathcal{O}_V} & V_U. \end{array}$$

Corollary 2 (Paired Fixed Points). *A point $p_\star$ is a fixed point of $\mathcal{O}_P$ if and only if*

$$v_\star = h(p_\star)$$

is a fixed point of $\mathcal{O}_V$.

The paired-update statement does not imply that a coordinate-dependent optimisation rule retains the same algebraic formula in both descriptions. For a nonlinear h , an ordinary Euclidean update such as

$$p_{k+1} = p_k - \eta \nabla L_P(p_k)$$

need not transform into

$$v_{k+1} = v_k - \eta \nabla L_V(v_k).$$

Exact correspondence applies to the conjugated update

$$\mathcal{O}_V = h \circ \mathcal{O}_P \circ h^{-1},$$

or, for gradient dynamics, when the gradients are defined using the corresponding metrics related by

$$g_P = h^* g_V.$$

Accordingly, paired optimisation means that a physically admissible optimisation process and a semantic optimisation process describe the same local trajectory under the physical–semantic correspondence. As elsewhere, this claim is substantive only when the admissible objectives and optimisation rules on the two sides are independently specified.

8 Equivalence and Duality

The preceding construction preserves the direction of composition:

$$F_2 \circ F_1 \mapsto T_2 \circ T_1. \quad (46)$$

It is therefore an equivalence, not an order-reversing duality.

Definition 7 (Physical–Semantic Duality). A physical–semantic duality is a bijection

$$\mathcal{D} : \mathcal{M}_P \longrightarrow \mathcal{M}_V \quad (47)$$

satisfying

$$\mathcal{D}(F_2 \circ F_1) = \mathcal{D}(F_1) \circ \mathcal{D}(F_2). \quad (48)$$

Equation (48) defines an anti-isomorphism. Categorically, one seeks

$$\mathbf{Phys} \simeq \mathbf{Sem}^{op}. \quad (49)$$

The local vectorisation h alone does not produce (48).

Suppose instead that the semantic transformation structure admits an anti-automorphism

$$\star : \mathcal{M}_V \longrightarrow \mathcal{M}_V \quad (50)$$

with

$$(T_2 \circ T_1)^\star = T_1^\star \circ T_2^\star. \quad (51)$$

Then define

$$\mathcal{D} := \star \circ \mathcal{C}_h. \quad (52)$$

Lemma 3 (Candidate Duality). If $\star$ preserves the admissible semantic transformation collection, then

$$\mathcal{D}(F_2 \circ F_1) = \mathcal{D}(F_1) \circ \mathcal{D}(F_2). \quad (53)$$

Proof. Apply Lemma 2 followed by (51).

An adjoint, time-reversal map, or another physically instantiated order-reversing operation may provide such a $\star$. No particular choice is implied by the local semantic vectorisation itself.

8.1 Structural Transport Through the Hylogical Map

The local hylogical vectorisation does not imply that every physical property automatically becomes a semantic property. Transfer is restricted to the operations, relations, and observables preserved by the map. The following definitions make this restriction explicit.

Definition 3 (Retained Structural Signature). *Let*

$$\mathfrak{P} := (P_U, \mathcal{S}_P), \quad \mathfrak{V} := (V_U, \mathcal{S}_V), \quad (6)$$

where the retained physical structure $\mathcal{S}_P$ consists of families of operations, relations, and observables of the form

$$F_P^\alpha : P_U^{k_\alpha} \longrightarrow P_U, \quad (7)$$

$$R_P^\beta \subseteq P_U^{\ell_\beta}, \quad (8)$$

$$f_P^\gamma : P_U^{m_\gamma} \longrightarrow Z_\gamma. \quad (9)$$

Here, each Z_γ is a fixed auxiliary codomain. Let

$$F_V^\alpha, \quad R_V^\beta, \quad f_V^\gamma \quad (10)$$

denote the corresponding semantic operations, relations, and observables on V_U .

Definition 4 (Hylogical Structural Compatibility). Let

$$h : P_U \longrightarrow V_U \quad (11)$$

be the local physical-semantic vectorisation. The retained physical and semantic structures are said to be h -compatible when, for every admissible tuple $(p_1, \dots, p_k)$, the following conditions hold:

$$h(F_P^\alpha(p_1, \dots, p_{k_\alpha})) = F_V^\alpha(h(p_1), \dots, h(p_{k_\alpha})), \quad (12)$$

$$(p_1, \dots, p_{\ell_\beta}) \in R_P^\beta \iff (h(p_1), \dots, h(p_{\ell_\beta})) \in R_V^\beta, \quad (13)$$

$$f_P^\gamma(p_1, \dots, p_{m_\gamma}) = f_V^\gamma(h(p_1), \dots, h(p_{m_\gamma})). \quad (14)$$

Thus h is an isomorphism with respect to the retained structure, not merely a bijection between the underlying sets.

[Local Hylogical Structural Transfer] Suppose that

$$h : P_U \longrightarrow V_U \quad (15)$$

is a bijection and that $\mathcal{S}_P$ and $\mathcal{S}_V$ are h -compatible.

Let

$$\varphi_P(p_1, \dots, p_r) \quad (16)$$

be any statement constructed solely from the retained physical operations, relations, observables, equality, logical connectives, and quantification over P_U . Let

$$\varphi_V(v_1, \dots, v_r) \quad (17)$$

be the corresponding semantic statement obtained by replacing each physical object in φ_P with its h -compatible semantic counterpart.

Then

$$\mathfrak{P} \models \varphi_P(p_1, \dots, p_r) \iff \mathfrak{V} \models \varphi_V(h(p_1), \dots, h(p_r)). \quad (18)$$

Proof. First consider a term constructed from the retained operations. Repeated application of (12) shows that evaluating the term and then applying h gives the same result as first applying h to its arguments and then evaluating the corresponding semantic term.

Atomic relational statements are preserved by (13), while equalities of retained observables are preserved by (14). Preservation under negation, conjunction, disjunction, and implication follows immediately.

Finally, suppose

$$\mathfrak{P} \models \exists x \in P_U \psi_P(x, p_1, \dots, p_r). \quad (19)$$

Let $x^* \in P_U$ be a witness. By the preceding steps,

$$\mathfrak{V} \models \psi_V(h(x^*), h(p_1), \dots, h(p_r)). \quad (20)$$

Hence $h(x^*)$ is a semantic witness. The converse follows by applying the same argument to h^{-1} . Universal quantification follows from bijectivity. Therefore every statement constructed from the retained structure is preserved in both directions. $\square$

Corollary 3 (Transfer of Existence and Uniqueness). *Under the hypotheses of the preceding proposition,*

$$\mathfrak{P} \models \exists! x \in P_U \psi_P(x, p) \quad (21)$$

if and only if

$$\mathfrak{V} \models \exists! v \in V_U \psi_V(v, h(p)). \quad (22)$$

If x^ is the unique physical witness, then*

$$v^* = h(x^*) \quad (23)$$

is the unique semantic witness.

Proof. Existence follows from structural transfer. If two distinct semantic witnesses existed, their distinct inverse images under the bijection h would be two physical witnesses, contradicting physical uniqueness. The converse follows through h^{-1} . $\square$

Corollary 4 (Transfer of Result Families). *Let I be an index set and let*

$$\{\varphi_{P,i}\}_{i \in I} \quad (24)$$

be a family of physical statements expressed entirely in the retained structure. Then

$$\mathfrak{P} \models \varphi_{P,i} \quad \text{for every } i \in I \quad (25)$$

if and only if

$$\mathfrak{V} \models \varphi_{V,i} \quad \text{for every } i \in I, \quad (26)$$

where $\varphi_{V,i}$ is the h -transported semantic statement.

Consequently, a compatible family of physical results determines a corresponding family of semantic results, and conversely.

Proof. Apply Proposition 18 separately to every $i \in I$. $\square$

[Concrete Transport Rules] The required compatibility relation depends on the type of structure being transported. Subject to the corresponding regularity conditions, common instances include

$$A_V := h(A_P), \quad (27)$$

$$F_V := h \circ F_P \circ h^{-1}, \quad (28)$$

$$X_V := h_* X_P, \quad (29)$$

$$\mu_V := h_{\#} \mu_P, \quad (30)$$

$$T_V := (h^{-1})^* T_P, \quad (31)$$

$$J_V := J_P \circ h^{-1}. \quad (32)$$

Here $A_P \subseteq P_U$ is a physical subset, F_P is a physical transformation, X_P is a vector field, μ_P is a probability measure, T_P is a covariant tensor, and J_P is a real-valued objective.

For example, whenever the maximisers exist,

$$h\left(\arg \max_{p \in P_U} J_P(p)\right) = \arg \max_{v \in V_U} J_V(v). \quad (33)$$

Likewise, if $\mu_V = h_{\#} \mu_P$, then every integrable semantic observable f_V satisfies

$$\int_{V_U} f_V(v) \mu_V(dv) = \int_{P_U} f_V(h(p)) \mu_P(dp). \quad (34)$$

[Structure-Specific Strength] A bijection transfers only set-theoretic structure. Transfer of topological results requires a homeomorphism; transfer of smooth results requires a diffeomorphism; transfer of probabilistic results requires a measurable isomorphism; transfer of dynamical results requires conjugate evolution; and transfer of metric results requires the corresponding pullback or isometry condition.

Accordingly, the existence of the hylogical map alone does not transport every property of the physical domain.

[Formal and Substantive Content] If the semantic structure $\mathcal{S}_V$ is defined entirely by transporting $\mathcal{S}_P$ through h , the structural-transfer proposition is a formal consistency result. Its substantive content arises when the physical and semantic structures are specified independently and are subsequently proved to satisfy (12)–(14).

[Locality and Quotient Limitation] The result applies only on the local domains

$$h : P_U \longrightarrow V_U. \quad (35)$$

It therefore transfers only statements whose relevant objects and trajectories remain inside those domains.

Moreover, if

$$q : \tilde{P} \longrightarrow P_U \quad (36)$$

is the preceding observational quotient, structural transfer recovers results about the observed equivalence class $q(\tilde{p})$, not about the omitted location of $\tilde{p}$ inside its fibre

$$q^{-1}(\{q(\tilde{p})\}). \quad (37)$$

No assumption-free recovery of discarded exterior or within-fibre information follows.

9 Conclusion

This research note extends local physical–semantic vectorisation from states to transformations. Quotient-compatible physical processes induce transformations on observed physical equivalence classes. Conjugation by the semantic vectorisation then relates these transformations to SVS calculations.

When the physically admissible transformations and semantic calculations are specified independently and satisfy

$$h\mathcal{M}_Ph^{-1} = \mathcal{M}_V, \quad (54)$$

the two descriptions have equivalent transformation structure. Their local tangent operators form isomorphic algebras, dynamical systems are conjugate, compatible metrics are isometric, and quotient-compatible scalar observables correspond.

The result remains local and quotient-relative. A semantic state recovers an observed physical equivalence class, not a unique complete physical state. Likewise, a semantic calculation has a physical realisation only when its corresponding quotient transformation descends from an admissible physical process.

Ordinary conjugation provides a physical–semantic equivalence. A stronger duality requires an additional order-reversing structure. Identifying whether physically instantiated observer–world systems naturally supply such a structure remains open.

References

- [1] P. Tohouri, “Approaching ORBIT: Examinations of Variant Agnostic Information Traversals in ML,” Aug. 2026. DOI: [10.6084/m9.figshare.32826848.v8](https://doi.org/10.6084/m9.figshare.32826848.v8). [Online]. Available: https://figshare.com/articles/preprint/7259230_/32826848.